

Final Report: DARE-KF; Kalman Filter–Based Dumbbell Satellite Attitude Estimation

Michael Deetz, Piyush More and Jack Winings

Abstract

This project focuses on estimating the attitude and controlling the orientation of a rigid dumbbell-shaped satellite. Attitude estimation is performed by fusing measurements from an onboard gyroscope, sun sensor, and magnetometer, while explicitly accounting for realistic sensor noise and time-varying bias drift. Two estimation frameworks are implemented and compared: a Multiplicative Extended Kalman Filter (MEKF) and an Unscented Kalman Filter (UKF). Because the UKF demonstrated superior estimation performance, it is selected to provide real-time state estimates to a Model Predictive Control (MPC) algorithm. This combined UKF-MPC architecture is then utilized to successfully command the satellite and maintain precise sun-pointing, demonstrating the effectiveness of the closed-loop system for advanced spacecraft attitude control.

I. INTRODUCTION

Accurate attitude estimation is a fundamental requirement for spacecraft operations, enabling precise pointing and autonomous control. For small satellites with constrained resources and low-cost sensors, robust estimation algorithms that account for realistic noise and time-varying gyroscope bias drift are essential to mission success.

This report presents the design and evaluation of an attitude estimation and control system for a rigid dumbbell-shaped satellite. Attitude is determined by fusing measurements from an onboard gyroscope, sun sensor, and magnetometer. Two nonlinear estimation frameworks are implemented and compared: a Multiplicative Extended Kalman Filter (MEKF) and an Unscented Kalman Filter (UKF) [4], [5]. Both filters estimate a six-dimensional error state consisting of the attitude error vector and gyroscope bias.

The UKF demonstrated superior estimation performance and is selected to provide real-time state estimates to a Model Predictive Control (MPC) algorithm. The combined UKF–MPC system is validated in closed-loop simulation over a full orbital period, successfully achieving and maintaining precise sun-pointing from a large initial misalignment.

II. MAIN BODY

SYSTEM DESCRIPTION

The spacecraft is modeled as a rigid dumbbell system consisting of two spheres connected by a thin rod. Both translational and rotational dynamics are considered.

EQUATIONS OF MOTION

The spacecraft dynamics are modeled using standard orbital mechanics and rigid body formulations [2], [3]. Translational motion follows the two-body equation:

$$\mathbf{a} = -\frac{\mu}{\|\mathbf{r}\|^3}\mathbf{r} \quad (1)$$

where $\mu = Gm_{\text{earth}}$. Rotational motion is governed by Euler’s equation and quaternion kinematics:

$$\dot{\boldsymbol{\omega}} = -\mathbf{I}^{-1}(\boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega}) \quad (2)$$

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega})\mathbf{q} \quad (3)$$

The authors are with Aeronautics and Astronautics Engineering, Purdue university, West Lafayette, IN, 47907 USA (e-mail: mjdeetz@purdue.edu, morep@purdue.edu, jwinings@purdue.edu)

DUMBBELL INERTIA CALCULATION

The inertia tensor is computed about the center of mass (midpoint of the rod) using the parallel axis theorem. For two spheres of mass m_s and radius R located at $\pm L/2$ along the x -axis, and a rod of mass m_r and effective length $(L - 2R)$, the diagonal components are:

$$I_{xx} = \frac{4}{5}m_s R^2 \quad (4)$$

$$I_{yy} = I_{zz} = 2 \left(\frac{2}{5}m_s R^2 + m_s \frac{L^2}{4} \right) + \frac{1}{12}m_r (L - 2R)^2 \quad (5)$$

giving the diagonal inertia tensor $\mathbf{I} = \text{diag}(I_{xx}, I_{yy}, I_{zz})$. The rod contributes nothing to I_{xx} since it is aligned along that axis. As a result, $I_{xx} \ll I_{yy} = I_{zz}$, making the x -axis the minimum-inertia spin axis of the dumbbell.

SENSOR MODELS

The attitude estimation system incorporates three sensors: a gyroscope, a sun sensor, and a magnetometer, each modeled with realistic noise characteristics representative of small satellite hardware.

Gyroscope

The measured angular rate is modeled as:

$$\boldsymbol{\omega}_{meas} = \boldsymbol{\omega}_{true} + \mathbf{b} + \mathbf{n}, \quad \mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I}), \quad \sigma_g = 0.01^\circ/\text{s} \quad (6)$$

where $\mathbf{b}$ is a time-varying bias and $\mathbf{n}$ is zero-mean Gaussian noise. The bias follows a first-order Gauss–Markov process:

$$\mathbf{b}_{k+1} = \alpha \mathbf{b}_k + \sqrt{1 - \alpha^2} \sigma_b \boldsymbol{\eta}_b, \quad \alpha = e^{-\Delta t/\tau} \quad (7)$$

with correlation time $\tau = 2000$ s and stability $\sigma_b = 0.5$ deg/hr. The bias is initialized as $\mathbf{b}_0 \sim \mathcal{N}(\mathbf{0}, (2 \text{ deg/hr})^2 \mathbf{I})$, representing an unknown cold-start bias that the filter must estimate.

Sun Sensor and Magnetometer

Both vector sensors share the same rotation-based noise model. The true body-frame reference vector is obtained via the attitude DCM:

$$\hat{\mathbf{v}}_b = \mathbf{C}_b^i \hat{\mathbf{v}}_i \quad (8)$$

A small-angle noise quaternion is formed from $\delta \boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$ and applied to yield the noisy measurement:

$$\mathbf{v}_{meas} = \frac{\mathbf{R}_n \hat{\mathbf{v}}_b}{\|\mathbf{R}_n \hat{\mathbf{v}}_b\|} \quad (9)$$

This preserves the unit-vector constraint exactly. The sun sensor uses $\hat{\mathbf{s}}_i = [1, 0, 0]^T$ with $\sigma_{ss} = 0.5^\circ$, and the magnetometer uses $\hat{\mathbf{m}}_i \propto [0.2, 0.3, -0.9]^T$ with $\sigma_{mag} = 1.0^\circ$.

The combined measurement vector and noise covariance delivered to the filters at each timestep are:

$$\mathbf{z} = \begin{bmatrix} \mathbf{s}_{meas} \\ \mathbf{m}_{meas} \end{bmatrix} \in \mathbb{R}^6, \quad \mathbf{R} = \begin{bmatrix} \sigma_{ss}^2 \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \sigma_{mag}^2 \mathbf{I}_3 \end{bmatrix} \quad (10)$$

MULTIPLICATIVE EXTENDED KALMAN FILTER - MKF

III. PROBLEM FORMULATION

A. Frames of Reference

Two reference frames are used:

- $\mathcal{I}$: Earth-Centered Inertial (ECI) frame
- $\mathcal{B}$: Spacecraft body frame

B. Attitude Representation

Attitude is represented using a unit quaternion:

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \mathbf{q}_v \end{bmatrix}, \quad q_0^2 + \|\mathbf{q}_v\|^2 = 1 \quad (11)$$

where q_0 is the scalar part and $\mathbf{q}_v \in \mathbb{R}^3$ is the vector part. A rotation of angle ϕ about unit axis $\hat{\mathbf{n}}$ is:

$$\mathbf{q} = \begin{bmatrix} \cos(\phi/2) \\ \hat{\mathbf{n}} \sin(\phi/2) \end{bmatrix} \quad (12)$$

The rotation matrix satisfying $\mathbf{v}^I = R(\mathbf{q})\mathbf{v}^B$ is:

$$R(\mathbf{q}) = (q_0^2 - \|\mathbf{q}_v\|^2)I + 2\mathbf{q}_v\mathbf{q}_v^T + 2q_0[\mathbf{q}_v \times] \quad (13)$$

where the skew-symmetric cross-product matrix is:

$$[\mathbf{a} \times] = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \quad (14)$$

Hamilton quaternion multiplication is:

$$\mathbf{p} \otimes \mathbf{q} = \begin{bmatrix} p_0q_0 - \mathbf{p}_v^T \mathbf{q}_v \\ p_0\mathbf{q}_v + q_0\mathbf{p}_v + \mathbf{p}_v \times \mathbf{q}_v \end{bmatrix} \quad (15)$$

The quaternion inverse (conjugate for unit quaternions) is $\mathbf{q}^{-1} = [q_0; -\mathbf{q}_v]$.

IV. MULTIPLICATIVE EXTENDED KALMAN FILTER

A. Motivation

The quaternion unit norm constraint means attitude lives on $S^3 \subset \mathbb{R}^4$, not in flat Euclidean space. A standard additive EKF update violates this constraint. The MEKF resolves this by factoring the true quaternion as:

$$\mathbf{q}_{\text{true}} = \delta\mathbf{q} \otimes \hat{\mathbf{q}} \quad (16)$$

where $\hat{\mathbf{q}}$ is a nominal quaternion propagated with full nonlinear kinematics, and $\delta\mathbf{q}$ represents a small attitude error estimated by the EKF.

B. Error State

The EKF error state is:

$$\mathbf{x} = \begin{bmatrix} \delta\boldsymbol{\theta} \\ \delta\mathbf{b} \end{bmatrix} \in \mathbb{R}^6 \quad (17)$$

where $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ is the attitude error rotation vector and $\delta\mathbf{b} \in \mathbb{R}^3$ is the bias error. The small-angle approximation gives:

$$\delta\mathbf{q} \approx \begin{bmatrix} 1 \\ \frac{1}{2}\delta\boldsymbol{\theta} \end{bmatrix} \quad (18)$$

By construction of the reset step, $E[\mathbf{x}] = \mathbf{0}$ at all times.

C. Propagation

The best estimate of angular velocity is:

$$\hat{\boldsymbol{\omega}} = \boldsymbol{\omega}_{meas} - \hat{\mathbf{b}} \quad (19)$$

The nominal quaternion is propagated via:

$$\dot{\hat{\mathbf{q}}} = \frac{1}{2} \Omega(\hat{\boldsymbol{\omega}}) \hat{\mathbf{q}} \quad (20)$$

The linearized form of the continuous-time error dynamics is:

$$F = \begin{bmatrix} -[\hat{\boldsymbol{\omega}} \times] & -I_3 \\ \mathbf{0}_3 & \mathbf{0}_3 \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (21)$$

The discrete state transition matrix (first-order approximation) is:

$$\Phi_k \approx I_6 + F \Delta t \quad (22)$$

Covariance propagation:

$$P_{k|k-1} = \Phi_k P_{k-1|k-1} \Phi_k^T + Q_k \quad (23)$$

where:

$$Q_k = \begin{bmatrix} \sigma_g^2 I & \mathbf{0} \\ \mathbf{0} & (1 - \alpha^2) \sigma_b^2 I \end{bmatrix} \Delta t \quad (24)$$

D. Measurement Update

The predicted body-frame measurements are:

$$\hat{\mathbf{s}}^B = R(\hat{\mathbf{q}})^T \mathbf{s}^I, \quad \hat{\mathbf{m}}^B = R(\hat{\mathbf{q}})^T \mathbf{m}^I \quad (25)$$

The combined innovation is:

$$\mathbf{r} = \begin{bmatrix} \mathbf{y}_{\text{sun}} - \hat{\mathbf{s}}^B \\ \mathbf{y}_{\text{mag}} - \hat{\mathbf{m}}^B \end{bmatrix} \in \mathbb{R}^{6 \times 6} \quad (26)$$

Linearizing the measurement model about $\delta \boldsymbol{\theta} = \mathbf{0}$ gives:

$$\frac{\partial \hat{\mathbf{y}}}{\partial \delta \boldsymbol{\theta}} = -[\hat{\mathbf{y}}^B \times] \quad (27)$$

Bias is not directly measured, so the full 6×6 measurement Jacobian has zeros in the three rightmost columns. It is as follows:

$$H_k = \begin{bmatrix} -[\hat{\mathbf{s}}^B \times] & \mathbf{0}_3 \\ -[\hat{\mathbf{m}}^B \times] & \mathbf{0}_3 \end{bmatrix} \quad (28)$$

The Kalman gain is:

$$K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \quad (29)$$

where:

$$R_k = \begin{bmatrix} \sigma_{\text{sun}}^2 I_3 & \mathbf{0} \\ \mathbf{0} & \sigma_{\text{mag}}^2 I_3 \end{bmatrix} \quad (30)$$

The error state update is:

$$\mathbf{x}^+ = K_k \mathbf{r}_k = \begin{bmatrix} \delta \boldsymbol{\theta}^+ \\ \delta \mathbf{b}^+ \end{bmatrix} \quad (31)$$

Covariance update (Joseph form for numerical stability):

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} (I - K_k H_k)^T + K_k R_k K_k^T \quad (32)$$

E. Reset Step

After the update, corrections are applied to the nominal state and the error state is zeroed:

$$\delta \mathbf{q} = \frac{1}{\left\| \begin{bmatrix} 1 \\ 1 \\ \frac{1}{2} \delta \boldsymbol{\theta}^+ \end{bmatrix} \right\|} \begin{bmatrix} 1 \\ 1 \\ \frac{1}{2} \delta \boldsymbol{\theta}^+ \end{bmatrix}, \quad \hat{\mathbf{q}}^+ = \delta \mathbf{q} \otimes \hat{\mathbf{q}}^- \quad (33)$$

$$\hat{\mathbf{b}}^+ = \hat{\mathbf{b}}^- + \delta \mathbf{b}^+ \quad (34)$$

$$\delta \boldsymbol{\theta} \leftarrow \mathbf{0}, \quad \delta \mathbf{b} \leftarrow \mathbf{0} \quad (35)$$

This reset maintains $E[\mathbf{x}] = \mathbf{0}$ and must be performed before the next propagation step.

UNSCENTED KALMAN FILTER - UKF

The UKF is set up to estimate the same problem as the MEKF and uses the same error state so this section will begin with the UKF unique structure.

Sigma Points

Scaled Unscented Transform Parameters: The scaled unscented transform with tuning parameters α , β , and κ :

$$\lambda = \alpha^2(n + \kappa) - n, \quad \gamma = \sqrt{n + \lambda}, \quad n = 6. \quad (36)$$

The $\beta = 2$ choice is optimal for Gaussian priors [5]. The resulting $2n + 1 = 13$ sigma-point weights are:

$$W_m^{(0)} = \frac{\lambda}{n + \lambda}, \quad W_c^{(0)} = W_m^{(0)} + (1 - \alpha^2 + \beta), \quad (37)$$

$$W_m^{(i)} = W_c^{(i)} = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n. \quad (38)$$

The mean and covariance weights differ for $i = 0$ because the $(1 - \alpha^2 + \beta)$ term accounts for higher-order moment corrections in the covariance estimate.

Constructing the Sigma Points: Compute the lower Cholesky factor $\mathbf{S}_k$ such that $\mathbf{P}_k = \mathbf{S}_k \mathbf{S}_k^T$. The sigma points in error-state space are

$$\boldsymbol{\Xi}^{(0)} = \mathbf{0}, \quad \boldsymbol{\Xi}^{(i)} = +\gamma [\mathbf{S}_k]_{:,i}, \quad \boldsymbol{\Xi}^{(n+i)} = -\gamma [\mathbf{S}_k]_{:,i}, \quad i = 1, \dots, n. \quad (39)$$

Each point $\boldsymbol{\Xi}^{(i)} = [\delta \boldsymbol{\theta}^{(i)T}, \delta \mathbf{b}^{(i)T}]^T$ is converted to full-state form:

$$\mathbf{q}_\sigma^{(i)} = \delta \mathbf{q}(\delta \boldsymbol{\theta}^{(i)}) \otimes \hat{\mathbf{q}}_k, \quad \mathbf{b}_\sigma^{(i)} = \hat{\mathbf{b}}_k + \delta \mathbf{b}^{(i)}. \quad (40)$$

There is no $\delta \boldsymbol{\omega}$ component because angular velocity is not an estimated state.

Propagation

Per-Sigma-Point Propagation: Because each sigma point carries its own bias perturbation, the corrected angular rate is different for each point:

$$\boldsymbol{\omega}_c^{(i)} = \tilde{\boldsymbol{\omega}}_k - \mathbf{b}_\sigma^{(i)}. \quad (41)$$

This is the critical coupling: the bias sigma-point spread propagates into attitude uncertainty through (41), which then affects the predicted quaternion. If the bias spread did not enter here, the cross-covariance $\mathbf{P}_{xz}$ between the attitude error and the measurements would be severely underestimated.

Each quaternion sigma point is propagated by Euler integration of (3) and renormalized:

$$\mathbf{q}_p^{(i)} = \frac{\mathbf{q}_\sigma^{(i)} + \frac{\Delta t}{2} \Omega(\boldsymbol{\omega}_c^{(i)}) \mathbf{q}_\sigma^{(i)}}{\|\dots\|}. \quad (42)$$

The bias sigma points are propagated according to the Gauss–Markov model (7):

$$\mathbf{b}_p^{(i)} = \alpha \mathbf{b}_\sigma^{(i)}. \quad (43)$$

Reconstructing the Predicted Mean: The bias mean is a standard weighted sum:

$$\hat{\mathbf{b}}^- = \sum_{i=0}^{2n} W_m^{(i)} \mathbf{b}_p^{(i)}. \quad (44)$$

The quaternion mean cannot be computed by a simple weighted sum because quaternion space is not linear. We use the zeroth sigma point as a reference and average the attitude errors of all points relative to it:

$$\overline{\delta\boldsymbol{\theta}} = \sum_{i=0}^{2n} W_m^{(i)} \cdot 2 \left[\mathbf{q}_p^{(i)} \otimes \left(\mathbf{q}_p^{(0)} \right)^{-1} \right]_{2:4}, \quad (45)$$

then compose this mean error back onto the reference:

$$\hat{\mathbf{q}}^- = \delta\mathbf{q}(\overline{\delta\boldsymbol{\theta}}) \otimes \mathbf{q}_p^{(0)}, \quad (\text{renormalized}). \quad (46)$$

Before extracting the vector part in (45), quaternions with a negative scalar component are sign-flipped to ensure consistency during averaging.

Predicted Covariance: The attitude error of each propagated sigma point relative to the predicted mean is

$$\delta\boldsymbol{\theta}_p^{(i)} = 2 \left[\mathbf{q}_p^{(i)} \otimes (\hat{\mathbf{q}}^-)^{-1} \right]_{2:4}, \quad (47)$$

giving the 6D deviation vector $\delta\mathbf{x}_p^{(i)} = [\delta\boldsymbol{\theta}_p^{(i)T}, (\mathbf{b}_p^{(i)} - \hat{\mathbf{b}}^-)^T]^T$. The predicted covariance is

$$\mathbf{P}^- = \sum_{i=0}^{2n} W_c^{(i)} \delta\mathbf{x}_p^{(i)} \left(\delta\mathbf{x}_p^{(i)} \right)^T + \mathbf{Q}, \quad (48)$$

where the process noise covariance is

$$\mathbf{Q} = \text{diag}(\sigma_\tau^2 \mathbf{I}_3, \sigma_{b,p}^2 \mathbf{I}_3). \quad (49)$$

The attitude process noise σ_τ accounts for unmodeled disturbance torques. The bias process noise $\sigma_{b,p}$ is set consistent with the Gauss–Markov model in (7). Both may be inflated during tuning to prevent the filter from becoming overconfident before bias convergence.

Measurement Update

Predicted Measurements: For each propagated sigma point, the known inertial reference vectors are rotated into the predicted body frame:

$$\mathbf{z}_p^{(i)} = \begin{bmatrix} \mathbf{R}(\mathbf{q}_p^{(i)}) \hat{\mathbf{s}}_I \\ \mathbf{R}(\mathbf{q}_p^{(i)}) \hat{\mathbf{m}}_I \end{bmatrix} \in \mathbb{R}^6, \quad \hat{\mathbf{z}}^- = \sum_{i=0}^{2n} W_m^{(i)} \mathbf{z}_p^{(i)}. \quad (50)$$

Innovation Covariance and Cross-Covariance: Letting $\delta\mathbf{z}^{(i)} = \mathbf{z}_p^{(i)} - \hat{\mathbf{z}}^-$, the innovation covariance and state-measurement cross-covariance are:

$$\mathbf{S} = \sum_{i=0}^{2n} W_c^{(i)} \delta\mathbf{z}^{(i)} \left(\delta\mathbf{z}^{(i)} \right)^T + \mathbf{R}_{\text{meas}}, \quad (51)$$

$$\mathbf{P}_{xz} = \sum_{i=0}^{2n} W_c^{(i)} \delta\mathbf{x}_p^{(i)} \left(\delta\mathbf{z}^{(i)} \right)^T, \quad (52)$$

where $\mathbf{R}_{\text{meas}} = \text{diag}(\sigma_s^2 \mathbf{I}_3, \sigma_m^2 \mathbf{I}_3)$ is the measurement noise covariance. Note that $\mathbf{P}_{xz} \in \mathbb{R}^{6 \times 6}$ because the error state is 6-dimensional. The Kalman gain is

$$\mathbf{K} = \mathbf{P}_{xz} \mathbf{S}^{-1} \in \mathbb{R}^{6 \times 6}. \quad (53)$$

State and Covariance Update: The error-state correction is $\delta \mathbf{x}^* = \mathbf{K}(\mathbf{z}_{k+1} - \hat{\mathbf{z}}^-)$. Splitting $\delta \mathbf{x}^* = [\delta \boldsymbol{\theta}^{*T}, \delta \mathbf{b}^{*T}]^T$, the estimates update as:

$$\hat{\mathbf{q}}_{k+1} = \delta \mathbf{q}(\delta \boldsymbol{\theta}^*) \otimes \hat{\mathbf{q}}^- \quad (\text{renormalized}), \quad (54)$$

$$\hat{\mathbf{b}}_{k+1} = \hat{\mathbf{b}}^- + \delta \mathbf{b}^*. \quad (55)$$

The angular velocity estimate is then *derived*, not corrected by the gain:

$$\hat{\boldsymbol{\omega}}_{k+1} = \tilde{\boldsymbol{\omega}}_{k+1} - \hat{\mathbf{b}}_{k+1}. \quad (56)$$

This reflects the fact that angular velocity observability comes entirely from the gyro; the attitude sensors contribute only through their indirect improvement of the bias estimate. The covariance update with symmetrization is:

$$\mathbf{P}_{k+1} = \frac{1}{2} \left[(\mathbf{P}^- - \mathbf{K} \mathbf{S} \mathbf{K}^T) + (\mathbf{P}^- - \mathbf{K} \mathbf{S} \mathbf{K}^T)^T \right]. \quad (57)$$

The symmetrization step prevents floating-point rounding from making $\mathbf{P}$ slightly non-symmetric over many iterations.

SIMULATION RESULTS

First, we must compare the performance of both the UKF and MEKF to decide which filter should be used in conjunction with the MPC algorithm. Both the UKF and MEKF have similar attitude error magnitudes throughout the simulation as seen below. The UKF and MEKF both have sigma bounds which struggle to converge. When

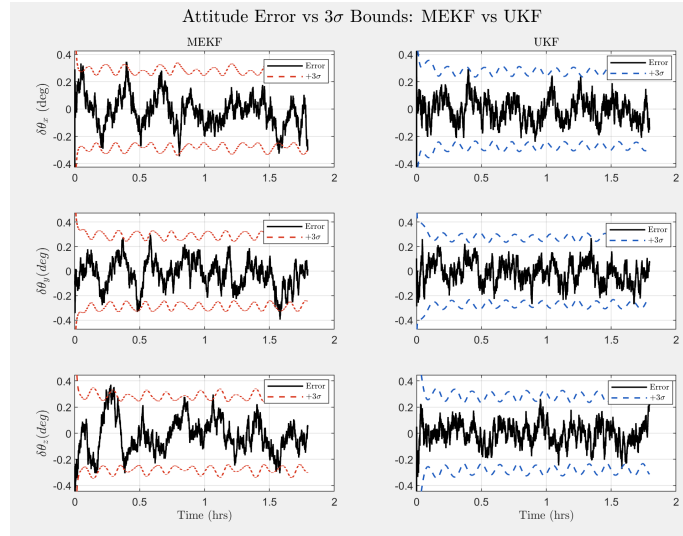


Fig. 1: MEKF and UKF Attitude Estimation Error

looking at how the filters perform in tracking the angular rates, they are both very similar. Note that the angular rate in the x direction is more greatly impacted by random torques due to that axis having the minimum moment of inertia.

In this comparison, the UKF and MEKF both receive the same sensor measurements, so the difference in angular rate estimate comes solely from the bias tracking. If $\sigma_b = 0.5$ deg/hr, we get the following graphs:

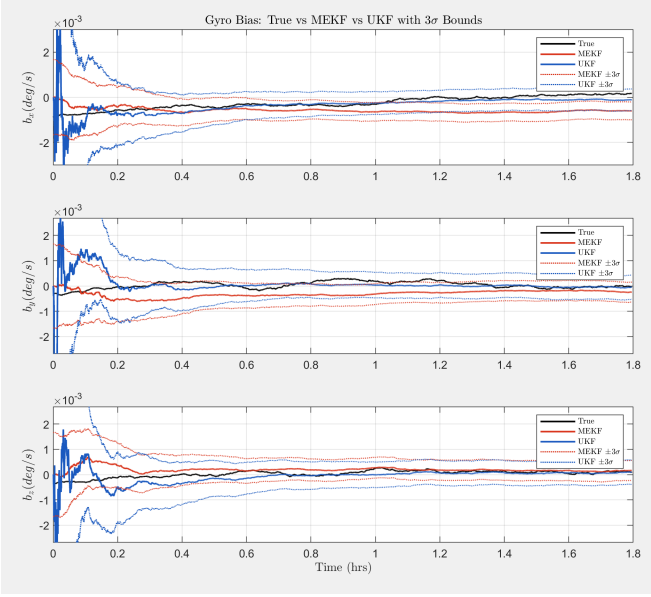


Fig. 2: MEKF and UKF bias estimation vs True bias

The two filters appear to track the angular rate very similarly. Neither filter has the angular rate in the state and it is thus estimated using the bias (41). This means each filter estimates the rates based on how well they estimate the bias and thus the UKF very slightly outperforms the MEKF. This becomes more noticeable When we push σ_b to a higher, unrealistic, 5 deg/hr.

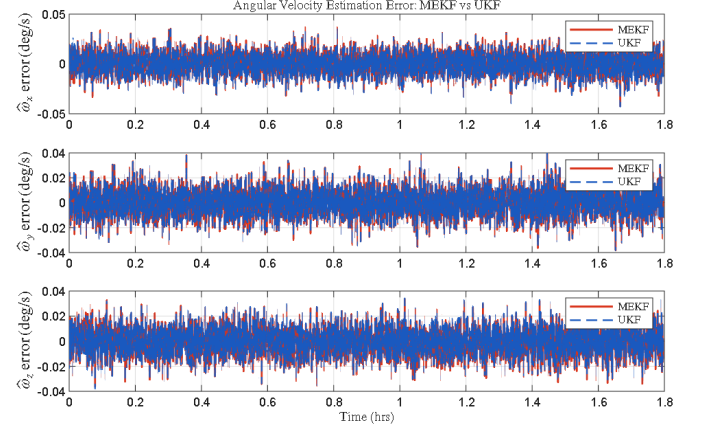


Fig. 3: MEKF and UKF Angular Velocity Estimation Error

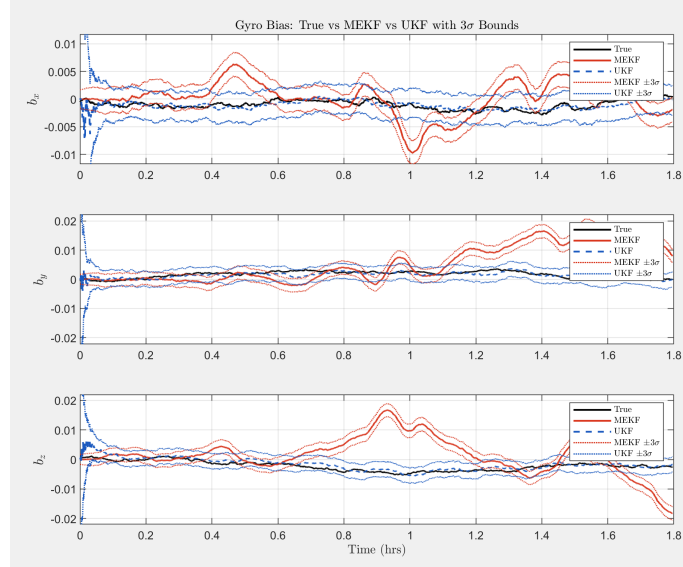


Fig. 4: MEKF and UKF bias estimation $\sigma_b = 5$ deg/hr

The UKF outperforms the MEKF, so it will be the filter that is utilized in the MPC algorithm.

V. MODEL PREDICTIVE CONTROL

Having established the UKF estimator, we now describe the MPC layer that closes the loop [6]. The UKF provides a real-time error-state estimate which the MPC uses to compute optimal control torques subject to actuator constraints.

A. Linearized Discrete-Time Model

The MPC operates on a linearized error-state representation. Rather than the full nonlinear quaternion dynamics used by the UKF, the controller state is defined as:

$$\mathbf{x} = \begin{bmatrix} \mathbf{q}_v \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^6, \quad (58)$$

where $\mathbf{q}_v = [q_1, q_2, q_3]^T$ is the vector part of the quaternion from the UKF. The scalar part q_0 is excluded since the MPC linearizes about the identity quaternion $\mathbf{q}^* = [1, 0, 0, 0]^T$, near which $q_0 \approx 1$ carries no independent information. With sampling period $T_s = 1$ s, the discrete-time matrices are:

$$A_d = \begin{bmatrix} I_3 & \frac{T_s}{2} I_3 \\ \mathbf{0}_3 & I_3 \end{bmatrix}, \quad B_d = \begin{bmatrix} \mathbf{0}_3 \\ T_s I^{-1} \end{bmatrix}, \quad (59)$$

and the reference state is $\mathbf{x}_{\text{ref}} = \mathbf{0}_6$, corresponding to sun-pointing attitude stabilization, where $\mathbf{q}_v = \mathbf{0}$ aligns the body x-axis with the inertial sun vector $\hat{s}_I = [1, 0, 0]^T$ and $\boldsymbol{\omega} = \mathbf{0}$ drives the angular rates to zero.

B. Batch Prediction and QP Formulation

Over a prediction horizon $N = 10$, the future state trajectory is stacked into the batch prediction following [6]:

$$\mathbf{X} = \mathbf{H} \mathbf{x}_k + \mathbf{G} \mathbf{U}, \quad (60)$$

where $\mathbf{U} = [\boldsymbol{\tau}_0^T, \dots, \boldsymbol{\tau}_{N-1}^T]^T \in \mathbb{R}^{3N}$ is the stacked control sequence, and:

$$\mathbf{H} = \begin{bmatrix} A_d \\ A_d^2 \\ \vdots \\ A_d^N \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} B_d & 0 & \cdots & 0 \\ A_d B_d & B_d & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ A_d^{N-1} B_d & \cdots & A_d B_d & B_d \end{bmatrix}. \quad (61)$$

The MPC minimizes the finite-horizon cost [6]

$$J = \sum_{k=0}^{N-1} \mathbf{x}_k^T Q \mathbf{x}_k + \boldsymbol{\tau}_k^T R \boldsymbol{\tau}_k + \mathbf{x}_N^T P \mathbf{x}_N, \quad (62)$$

with $Q = 0.05 I_6$, $R = 50 I_3$, and $P = Q$.¹ Substituting the batch prediction reduces this to the dense QP:

$$J = \mathbf{U}^T L \mathbf{U} + 2\mathbf{f}^T \mathbf{U} + \text{const}, \quad (63)$$

where $L = \mathbf{G}^T \bar{Q} \mathbf{G} + \bar{R}$ and $\mathbf{f} = \mathbf{G}^T \bar{Q} \mathbf{H}(\hat{\mathbf{x}}_k - \mathbf{x}_{\text{ref}})$, with $\bar{Q} = \text{blkdiag}(Q, \dots, Q, P)$ and $\bar{R} = \text{blkdiag}(R, \dots, R)$.

Physical actuator limits impose a symmetric box constraint enforced over the full horizon:

$$|\tau_i| \leq \tau_{\max} = 5 \times 10^{-4} \text{ N m}, \quad i = x, y, z, \quad (64)$$

written compactly as $E \mathbf{U} \leq \mathbf{W}$, with $E = [I_{3N}; -I_{3N}]$ and $\mathbf{W} = \tau_{\max} \mathbf{1}_{6N}$. The QP is solved at each step using `mpcActiveSetSolver` [6], with the active constraint set updated at each iteration.

¹ Q and R here denote MPC penalty weights, distinct from the process and measurement noise covariances of the same name in Section IV.

C. UKF-MPC Closed-Loop Architecture

The combined system operates as a three-step loop at each time instant k :

$$\hat{\mathbf{x}}_k \xrightarrow{\text{MPC}} \boldsymbol{\tau}_k \xrightarrow{\text{Plant}} \mathbf{x}_{k+1} \xrightarrow{\text{UKF}} \hat{\mathbf{x}}_{k+1}. \quad (65)$$

At each step, the UKF delivers the error-state estimate $\hat{\mathbf{x}}_k = [\hat{\mathbf{q}}_{v,k}^T, (\tilde{\boldsymbol{\omega}}_k - \hat{\mathbf{b}}_k)^T]^T$ to the MPC. Only the first element of the optimal control sequence is applied (receding horizon principle):

$$\boldsymbol{\tau}_k = \mathcal{U}_{0|k}^*. \quad (66)$$

The true 13-state plant $[r^T, v^T, q^T, \omega^T]^T$ is then propagated via RK4 with actuator noise $\eta_\tau \sim \mathcal{N}(0, \sigma_\tau^2 I_3)$, $\sigma_\tau = 10^{-6}$ N m, and the quaternion is renormalized after each step. Fresh gyro, sun, and magnetometer measurements are generated from the propagated true state and fed back into the UKF as described in Section II.

VI. SIMULATION RESULTS

The closed-loop UKF-MPC system was simulated over one full orbital period (≈ 1.8 hrs) from a large initial misalignment of Roll $\phi_0 = 80^\circ$, Pitch $\theta_0 = 120^\circ$, Yaw $\psi_0 = 160^\circ$ with initial angular rates $[\omega_x, \omega_y, \omega_z] = [0.5, 0.3, 0.2]$ deg/s.

Attitude Convergence. Fig. 5 shows the angle between the body x-axis and the inertial sun vector. Starting from $\approx 65^\circ$, the MPC drives the satellite to within 1° of sun-pointing in under 0.3 hours and maintains it for the remainder of the orbit.

Control Effort. All three torque channels saturate at $\pm\tau_{\max} = 5 \times 10^{-4}$ N·m during the initial reorientation (Fig. 6), demonstrating full actuator utilization during the transient. After convergence, torques settle well within bounds throughout the remainder of the orbit.

UKF Performance. Fig. 7 shows the attitude estimation error alongside the filter's 3σ bounds. After a brief initial transient of $\approx 3^\circ$, all three axes settle within $\pm 0.5^\circ$ and remain bounded by their 3σ envelopes for the full orbital period, confirming filter consistency under closed-loop operation.

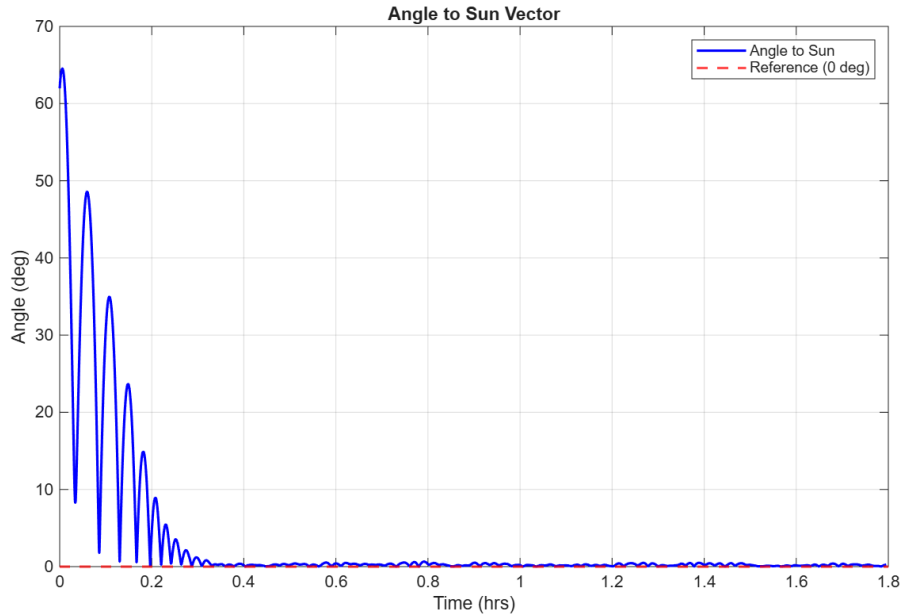


Fig. 5: Angle between body x-axis and inertial sun vector over one orbital period.

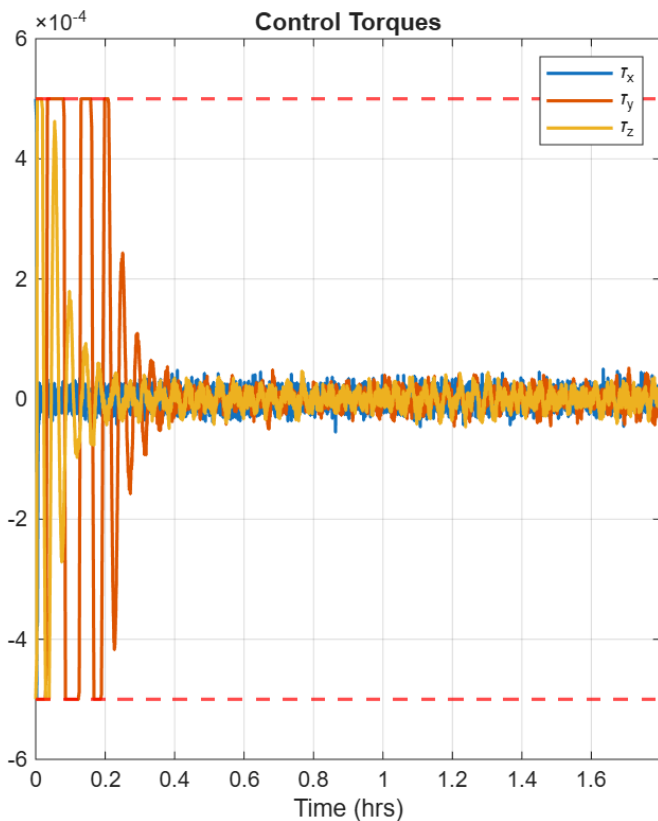


Fig. 6: MPC control torques over one orbital period. Torques saturate at $\pm\tau_{\max}$ during initial reorientation (0–0.3 hrs) and settle well within bounds thereafter.

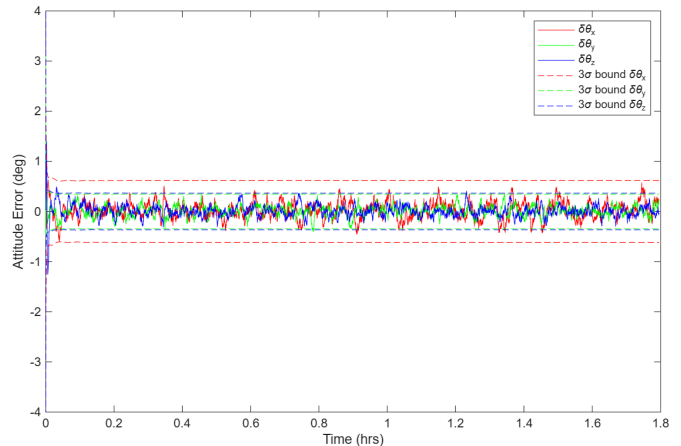


Fig. 7: UKF attitude error with 3σ bounds.

VII. CONCLUSION

This report presented the design and closed-loop evaluation of a UKF–MPC architecture for attitude estimation and control of a dumbbell-shaped satellite. Two estimation frameworks were developed and compared: an MEKF and a UKF, both operating on a six-dimensional error state consisting of attitude error and gyroscope bias. The UKF demonstrated superior consistency and bias tracking performance, motivating its selection as the estimator for the closed-loop system.

The integrated UKF–MPC system was simulated over a full orbital period from a large initial misalignment of 80, 120, and 160 in roll, pitch, and yaw respectively. The MPC successfully drove the satellite to within 1 of the sun-pointing target in under 0.3 hours, with all actuator torques remaining within the $\pm 5 \times 10^{-4}$ N·m saturation limit. UKF attitude errors settled within ± 0.5 and remained bounded by their 3σ envelopes for the full simulation, confirming filter consistency under closed-loop operation.

These results demonstrate that the UKF–MPC architecture is a viable and effective approach for spacecraft attitude control under realistic sensor noise and actuator constraints. Future work could explore adaptive noise tuning, higher-fidelity disturbance models, and hardware-in-the-loop validation.

VIII. CONTRIBUTIONS OF INDIVIDUAL MEMBERS

List your members and detail their contributions here.

1) Michael Deetz

Researched and created models for sensors. These gave us the "measured" data from the satellites orbit. Implemented a UKF to estimate the state of the spacecraft. Assisted with comparing the results of MEKF and UKF.

2) Piyush More

Developed and implemented the Model Predictive Control algorithm for attitude control. Formulated the batch prediction QP and integrated the MPC with the UKF to form the closed-loop architecture. Validated the combined system in MATLAB simulation and analyzed the closed-loop results.

3) Jack Winings

Assisted in modeling sensor data for the MEKF and UKF. Researched methods of estimating attitude. Utilized the MEKF to get around the issues regarding additive residuals in a typical EKF. Derived the necessary equations for the MEKF formulation. Worked on the comparison between both the MEKF and the UKF.

REFERENCES

- [1] Y. Bar-Shalom, T. Kirubarajan, and X Lin, “*Probabilistic Data Association Techniques for Target Tracking with Applications to Sonar, Radar and EO Sensors*,” IEEE Aerospace and Electronic Systems Magazine, vol. 20, no. 8, pp. 37-56, 2005.
- [2] A. Kutlu, “Design of Kalman Filter Based Attitude Determination Algorithms for a LEO Satellite and for a Satellite Attitude Control Test Setup,” M.S. thesis, Dept. Aersp. Eng., Middle East Technical Univ., Ankara, Turkey, Sept. 2008.
- [3] E. M. Keil, “Kalman Filter Implementation to Determine Orbit and Attitude of a Satellite in a Molniya Orbit,” M.S. thesis, Dept. Aersp. Ocean Eng., Virginia Polytechnic Institute and State Univ., Blacksburg, VA, USA, May 2014.
- [4] F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*. Springer, 2014.
- [5] D. Simon, *Optimal State Estimation: Kalman, H_∞ , and Nonlinear Approaches*. Wiley, 2006.
- [6] I. Hwang, “AAE 568: Applied Optimal Control and Estimation — Lecture Notes,” School of Aeronautics and Astronautics, Purdue University, West Lafayette, IN, Spring 2026.
- [7] F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, Springer, New York, NY, 2014.
- [8] M. Deetz, P. More, J. Winings “DARE-KF: Kalman Filter Implementation,” GitHub repository, 2026. [Online]. Available: <https://github.com/MichaelDeetz/DARE-KF.git>